Introduction to Health Economics

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- How can we determine the demand for health care?
    - How do we measure health care?
        - Number of visits to a doctor.
    - Offer a price chart to the patients and ask them about the number of visits?
        - Any issue with this solution?
        - Biased sample
    - Look at the fees paid and number of visits made by randomly selected sample in a locality.
    - Patients with insurance will face lower price and without insurance will face higher price
        - Any issue with this solution?
        - Individuals with insurance would be systematically different from those who are not insured.

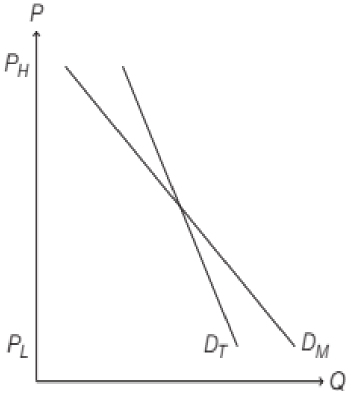

Figure 1: True demand, D_T, and measured demand, D_M, in a non-randomized study.

- Ideal Solution – Track the same sample of individuals in parallel universe of high prices and low prices.
  - Impossible !!!
  - Next best alternative - Randomized Controlled Trial
  - A study that assigns treatments randomly to different groups of study participants
- Two randomized health insurance experiments
  - RAND Health Insurance Experiment (HIE)
  - Oregon Medicaid Experiment (OME).

- RAND HIE
    - First large-scale randomized study in which insurance status was randomly assigned, and it is still the only such study ever conducted in the US.
    - Conducted between 1974 and 1982
    - Randomly assigned two thousand families from six American cities to one of several different health insurance plans for several years.
    - Plans varied on the generosity of coverage; in particular, the plans had different copayment rates.
        - Copayment rate: the fraction of the medical bill for which the patient is responsible.
        - People assigned to different plans had to pay different prices for the same services.
    - Four different plans: one plan with completely free care (0 percent copayment rate), and three other cost-sharing plans with 25 percent, 50 percent, and 95 percent copayments.
    - Limitation: Results of this study might not hold in the current context as health economy has changed in fundamental ways since the 1980s.

- Oregon Medicaid Experiment (OME)
  - The Oregon Health Plan (OHP) conceived in 1993—created by one of the first federal waivers of traditional Medicaid rules —consisted of two distinct programs from 2003 to 2013:
    - OHP Plus
    - OHP Standard
  - OHP Plus serves the categorically eligible Medicaid population, which includes children and pregnant women, the disabled, and families enrolled in Temporary Assistance to Needy Families (TANF). It is a full benefit package with no premiums.
  - OHP Standard, which is the program that was lotteried, is a Medicaid expansion program to cover low-income adults who are not categorically eligible for OHP Plus. Specifically, it covers adults aged 19–64 not otherwise eligible for public insurance who are Oregon residents, are U.S. citizens or legal immigrants, have been without health insurance for six months, have income below the federal poverty level (FPL), and have assets below USD 2,000. It provides relatively comprehensive benefits with no consumer cost sharing and monthly enrollee premiums range from USD 0 to USD 20 depending on income, with those below 10 percent of the FPL paying USD 0.

- In early 2002, about 110,000 people were enrolled in OHP Standard, about one-third the size of OHP Plus enrollment at that time
- Due to budgetary shortfalls, OHP Standard was closed to new enrollment from 2004 to 2007.
- In January 2008 the state reopened OHP Standard to new enrollment of an additional 10,000 adults.
- In the anticipation of excess demand, it was decied to add the new members through random lottery drawn from a new reservation list.
- The state conducted an extensive public awareness campaign about the lottery opportunity.
- From January 28 to February 29, 2008, anyone could be added to the lottery list by telephone, by fax, in person sign-up, by mail, or online.
- A total of 89,824 individuals were placed on the list during the five-week window it was open.

- The state conducted eight lottery drawings from the list with roughly equal numbers selected from each drawing; the drawings were fairly evenly spaced from March through September 2008.
- Selected individuals won the opportunity—for themselves and any household member (whether listed or not)—to apply for OHP Standard coverage.
- Treatment thus occurred at the household level. In total, 35,169 individuals—representing 29,664 households—were selected by lottery.
- If individuals in a selected household submitted the appropriate paperwork within 45 days after the state mailed them an application and demonstrated that they met the eligibility requirements, they were enrolled in OHP Standard.
- About 30 percent of selected individuals successfully enrolled. There were two main sources of slippage: about 60 percent of those selected sent back applications, and about half of those who sent back applications were deemed ineligible, primarily due to failure to meet the income requirement.

- Oregon Medicaid Experiment (OME) compared two groups of low-income adult Oregonians:
    - people who won a 2008 lottery to receive the opportunity to apply for public health insurance coverage through Medicaid
    - lottery entrants who did not win and were not given a chance to apply for Medicaid.
- Hence, the lottery winners tended to face lower out-of-pocket prices for care.

- Comparison between RAND HIE and OME
    - The RAND HIE studied a nationally representative population whereas the OME exclusively focuses on a low-income population.
    - The RAND HIE used a direct randomization of health insurance coverage, while the OME relied on a randomization scheme that was only indirectly related to insurance coverage (Lottery winners were only 25 percentage points more likely to be covered in the year following the lottery than the lottery losers were).
    - The OME included an uninsured group that was in part randomly assigned, while the RAND HIE did not include any participants who were totally without insurance.

- Outpatient care: any medical care that does not involve an overnight hospital stay. It is also sometimes called ambulatory care. E.g. runny noses, twisted ankles
- Inpatient care: medical care requiring overnight stays. E.g. Dengu, Asthma, Eye surgery
- Emergency Room care: medical care involving the emergency room stays. E.g. heart attacks

(a)			
	Avg # of annual episodes by condition		
Plan	Total	Acute	Chronic
Free	2.99	2.29	0.70
25%	2.32	1.78	0.54
50%	2.11	1.60	0.51
95%	1.90	1.44	0.46

(b)		
	% with visit	No. of visits
Lottery winners	63.6	2.22
Lottery losers	57.4	1.91

Figure 2: Evidence for outpatient care: (a) RAND HIE Study. (b) Oregon Medicaid Experiment.

- Both the RAND HIE and the OME find downward-sloping demand for outpatient care.

- What kind of price sensitivity to be expected for Inpatient care - High or Low ???

(a)	
Plan	Avg # of Annual Visits
Free	0.133
25%	0.109
50%	0.099
95%	0.098

(b)		
	% with visit	No. of visits
Lottery winners	7.4	0.103
Lottery losers	7.2	0.097

No significant difference at the $p = 10\%$ level.

Figure 3: Evidence for inpatient care: (a) RAND HIE Study. (b) OEM.

- The RAND HIE finds that demand is still downward-sloping but less elastic than demand for outpatient care.

- Price sensitivity for Emergency Room (ER) care.

(a)

Plan	Probability of ER use
Free	22%
25%	19%*
50%	20%
95%	15%**

* Indicates significantly different from the free plan at the $p = 5\%$ level.
** Indicates significantly different from the free plan at the $p = 1\%$ level.

(b)

	% with visit	No. of visits
Lottery winners	26.7	0.48
Lottery losers	26.1	0.47

No significant difference at the $p = 10\%$ level.

Figure 4: Evidence for ER care: (a) RAND HIE Study. (b) OME.

- Even for emergency room care – likely the most urgent kind – those on the highest copayment plan in the RAND HIE were less likely to buy care - Why?

- Pediatric care - Hard to imagine that parents would skip on care for their children because of price.

	0–6 years		7–16 years	
	Immunization	Any preventative	Immunization	Any preventative
Free	58.9	82.5	21.2	64.8
Copayment	48.7*	73.7*	21.7	59.6

* Statistically significant discrepancy from free plan.

Figure 5: Percentage with preventative pediatric care over three years, by age and care type.

- Demand curve for pediatric care (0-6 years) slopes downward.

- Mental health care

Plan	Mean expense ($)	Percentage of free plan
Free	42.2	–
25%	28.4	67%
50%	13.1	33%
95%	18.1	43%

Figure 6: Per-capita mental health expenditures, by plan type.

- Dental care

	Low-income group†		High-income group†	
	Percentage with any use	Average expenditures ($)	Percentage with any use	Average expenditures ($)
Free	57.8	317	74.7	339
95%	39.8*	216*	61.3*	234*

* Statistically significant discrepancy from free plan.
† The low-income group comprises the third of households with the lowest incomes. The high-income group comprises the third of households with the highest incomes.

Figure 7: Dental care utilization by income level.

- Prescription drugs

	No. of antibiotics per person	
Plan	Bacterial conditions	Viral conditions
Free	0.47	0.17
Copay	0.24**	0.08**

** Statistically significant discrepancy from the free plan.

Figure 8: Antibiotic use in the RAND HIE

- Non-randomized experiment evidence
- Citizens in U.S. are eligible for health insurance through Medicare when they turn 65 but not before
- If demand for health care is downward-sloping, we expect a jump in health care usage at age 65

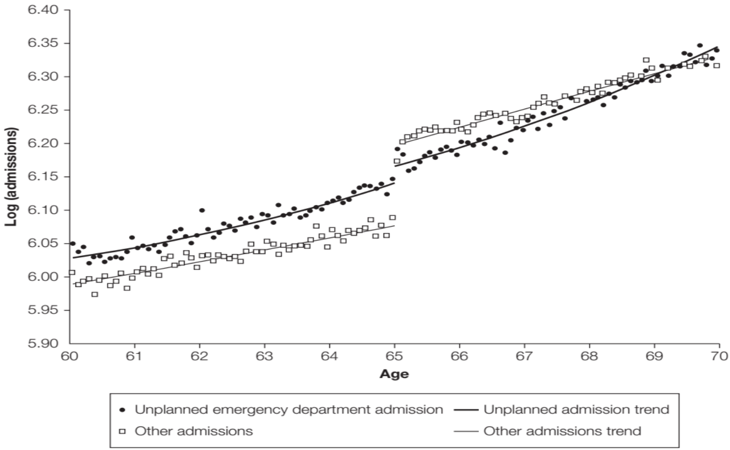

Figure 9: Emergency and non-emergency visits by age.

- Unplanned emergency department admissions follow a linear trend around the age of 65
- Other hospital admissions jump up at the age of 65

- How can we determine which type of demand is more price sensitive?

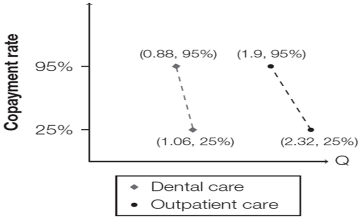

Figure 10: Data on outpatient and dental care.

- We use a measure to compare the relative price sensitivity of different goods.

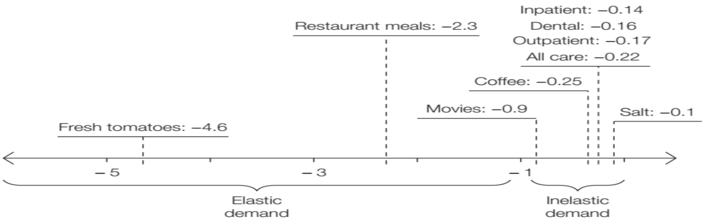

Figure 11: Price Elasticity.

- Does price for care affect health?

(a) Relative mortality rate		
	All participants	High-risk participants[a]
Free	0.99	1.90
Copay	1	2.10*

(b) Absolute two-year mortality rate	
	All participants
Lottery winners	0.8%
Lottery losers	0.8%

[a] Participants were classified as high risk based on their blood pressure, cholesterol levels, and smoking habits at the beginning of the study.
* Indicates significantly different from the free plan at the $p = 5\%$ level.

Figure 12: Evidence on mortality rates: (a) RAND HIE Study. (b) OEM.

- Mortality is an extreme outcome.

Condition	Free plan	Copay plan
FEV_1^a	95.0	94.8
Diastolic blood pressure (mm Hg)	78.0	78.8*
Cholesterol (mg/dl)	203	202
Glucose (mg/dl)	94.7	94.2
Abnormal thyroid level (% of sample)	2.4	1.7
Hemoglobin (g/100 ml)	14.5	14.5
Functional far vision (Snellen lines)	2.4	2.5*
Functional near vision (Snellen lines)	2.35	2.44*
Chronic joint symptoms (% of sample)	30.0	31.6

[a] FEV is forced expiratory volume in 1 second.
* Indicates significantly different from the free plan at the $p = 5\%$ level.

Figure 13: Health indicators by insurance plan in the RAND HIE.

	Lottery winners	Lottery losers
Survived one year after lottery	99.2%	99.2%
Self-reported health good	58.7%	54.8%**
Self-reported health fair or better	88.9%	86.0%**
Health about the same or better over last six months	74.7%	71.4%**
No. of days physical health good	22.2	21.9*
No. of days mental health good	19.3	18.7**
Did not screen positive for depression in last two weeks	69.4%	67.1%**

* Difference between lottery winners statistically significant at the $p = 5\%$ level.
** Difference between lottery winners statistically significant at the $p = 1\%$ level.

Figure 14: Effect of lottery win on health in the first year of the Oregon Medicaid Experiment.

- Conclusion
  - Demand curves for health care are downward sloping
  - BUT generally, price of health care does not seem to affect one's health
  - Exception is that price seems to affect the most vulnerable segments of the population (low-income, high blood pressure, etc.)

References

- Bhattacharya, J., Hyde, T., Tu, P. (2013). Health economics. Macmillan International Higher Education.